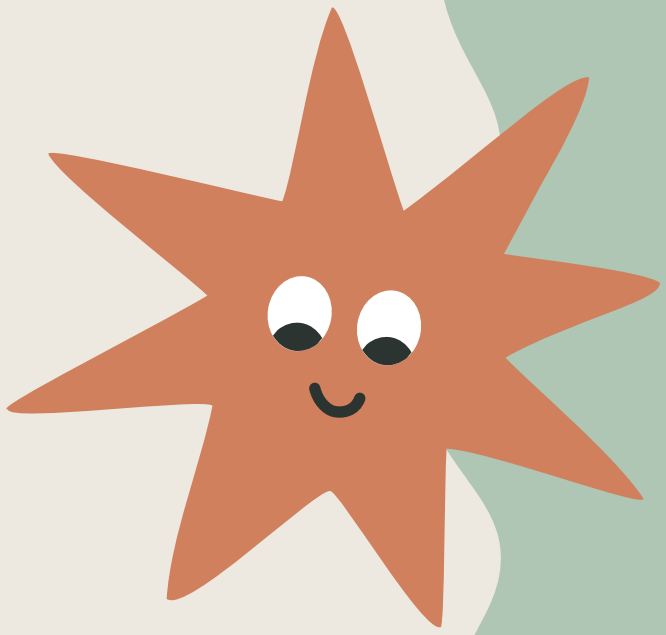
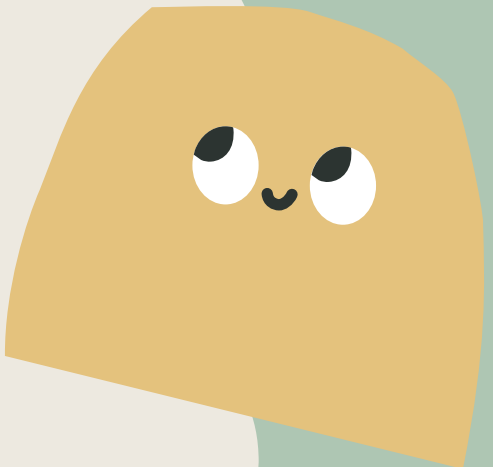
Introduction to Artificial Intelligence

Lecture 1





Learning Outcomes (Lecture 1)

➢At the end of this lecture, students will be able to:

- Define Artificial Intelligence.
- Explain the basic goals of AI.
- Differentiate AI from conventional computer programming.
- Identify characteristics of intelligent systems.
- Explain major areas/applications of AI.
- Differentiate Narrow AI and General AI at a conceptual level.
- Explain the relationship between AI, Machine Learning and Deep Learning.
- Identify AI systems in everyday life.
- Write a very simple Python example demonstrating an AI-like decision process.

Prepared By: Ehtisham Rasheed

What is Artificial Intelligence?

- The term has two parts:
- **Artificial** means something that is **created or produced by humans**, rather than occurring naturally
- For example:
  - Artificial light, artificial materials
- **Intelligence** is the ability to:
  - Learn, understand, reason, solve problems, make decisions, recognize patters and adopt to situations

Prepared By: Ehtisham Rasheed

What is Artificial Intelligence?

- Artificial Intelligence is the field of Computer Science concerned with creating systems that can perform tasks that normally require human-like intelligence
- Artificial Intelligence is the field of computer science concerned with developing systems that can perceive their environment, reason about information, learn from experience, and take actions to achieve specific goals.
- AI is the study and development of intelligent agents that perceive their environment and take actions that maximize the achievement of their goals.

Prepared By: Ehtisham Rasheed

What is Machine Learning?

- Machine learning is a branch of **Artificial Intelligence** that teaches computers to **learn from data** and **improve from experience** without being **explicitly programmed** for every single task
- In simple words:
- Machine Learning means teaching a computer through examples/data instead of writing every rule manually
- Machine Learning is one of the most widely used approaches for building modern AI systems

Prepared By: Ehtisham Rasheed

Is AI Really "Thinking"?

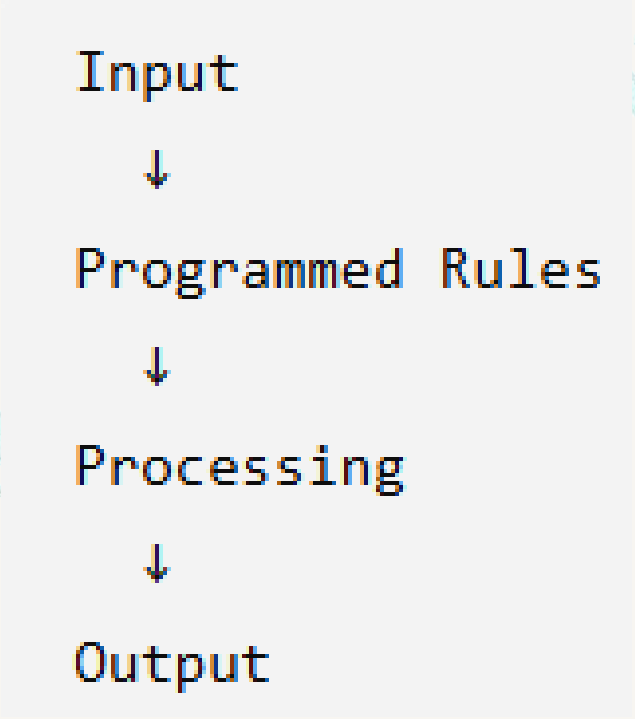
- When we say "The computer thinks.“
- we usually don't mean that the computer thinks exactly like a human
- Modern AI systems perform **computational processes** that can produce behavior that appears intelligent
- For example, a chess AI doesn't necessarily "understand chess" in the same way a human does
- It evaluates possible moves using algorithms and learned patterns

AI systems perform computational processes that can produce intelligent behavior

Prepared By: Ehtisham Rasheed

Why Do We Need AI?

- Traditional computer programs generally follow explicitly defined instructions



Prepared by Ibrahim Rasheed

Example Code

```
marks = 75


if marks >= 50:
    print("Pass")
else:
    print("Fail")
```

- The computer doesn't need to learn anything

Prepared by Rasheed

Question

- Given thousands of student records, predict which students are likely to fail next semester
- It is difficult to manually write rules for every possible situation
- AI allows computers to work with complex problems where:
    1. Rules may be difficult to define
    2. Data is large
    3. Environments change
    4. Uncertainty exists
    5. Decisions require patterns
    6. Learning from experience is useful

Prepared By: Ehtisham Rasheed

1. Rules May Be Difficult to Define (Face Recognition)

- Suppose we want a computer to recognize Ali's face
- It is extremely difficult to write rules like:
  - If the distance between the eyes is X, nose length is Y, face shape is Z → Ali
- Why? Because face can appear:
  - From different angles
  - With different lighting
  - With glasses
  - With different expressions
  - Partially covered
  - At different ages

Prepared By: Ehtisham Rasheed

1. Rules May Be Difficult to Define (Face Recognition)

- Instead of manually defining thousands of rules, AI/ML can **learn patterns from many example images**.
- Simple idea:
- Thousands of face images → AI model → Recognize a new face

Prepared By: Ehtisham Rasheed

2. Data is Large (YouTube Recommendations)

- Imagine YouTube has **millions of users watching billions of videos**
- For every user, there may be information about:
  - Videos watched
  - Watch duration
  - Likes
  - Searches
  - Subscriptions
  - Skipped videos
  - Viewing history

Prepared By: Ehtisham Rasheed

2. Data is Large (YouTube Recommendations)

- It would be extremely difficult for programmers to manually analyze all this data using fixed rules
- AI can process huge amounts of data and identify patterns such as:
  - Users who watched these videos are likely to be interested in these other videos.
- Simple idea:
- Huge amount of user data → AI → Personalized recommendations

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3. Environments Change (Self-Driving Car)

- Consider a car driving on a road
- The environment constantly changes:
  - Another car changes lane
  - A pedestrian crosses the road
  - Traffic lights change
  - Weather changes
  - An accident occurs
  - A new obstacle appears
  - Road condition change

Prepared By: Ehtisham Rasheed

3. Environments Change (Self-Driving Car)

- It is impossible to write a separate IF-ELSE rule for every possible situation
- An AI-based system can **perceive the current environment and make decisions based on what it observes**
- Simple idea:
- Sensors/Cameras → AI → Understand environment → Decide action

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4. Uncertainty exists (Medical Diagnosis)

- Suppose a patient has:
  - Fever
  - Cough
  - Fatigue
  - Headache
- These symptoms don't guarantee one particular disease
- Fever + cough ≠ definitely Disease X
- There is **uncertainty**

Prepared By: Ehtisham Rasheed

5. Decisions Require Patterns (Fraud Detection)

- A bank can have millions of transactions, one example:
  - PKR 50,000 transaction at 2:00 AM from an unusual location
- But that transaction alone doesn't necessarily mean fraud.
- AI can identify patterns involving:
  - Transaction amount
  - Location and/or Time
  - Previous transactions
  - Device
  - Customer behavior
  - Frequency of transactions

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5. Decisions Require Patterns (Fraud Detection)

- The system might recognize:
  - This transaction is very different from the customer's normal behavior
- It can then flag the transaction for investigation
- Simple idea:
- Transaction patterns → AI → Fraud probability → Alert

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6. Learning From Experience is Useful (Spam Detection)

- Initially, a spam filter may have difficulty recognizing new types of spam
- Suppose users repeatedly mark certain emails as spam
- The system can use these examples to improve its ability to recognize similar emails in the future

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6. Learning From Experience is Useful (Spam Detection)

➢For example:

```
Email 1 → Spam
Email 2 → Not Spam
Email 3 → Spam
Email 4 → Spam
        ↓
    Learning
        ↓
New Email → Spam / Not Spam
```

➢Simple idea:

➢The system improves its predictions based on previous examples

Prepared by Rasheed

AI vs Conventional Programming

Conventional programming

- The programmer provides:
  - Data + Rules → Program → Output
- For example:

```
Marks + Grading Rules
        ↓
     Program
        ↓
     Grade
```

Prepared by Tisham Rasheed

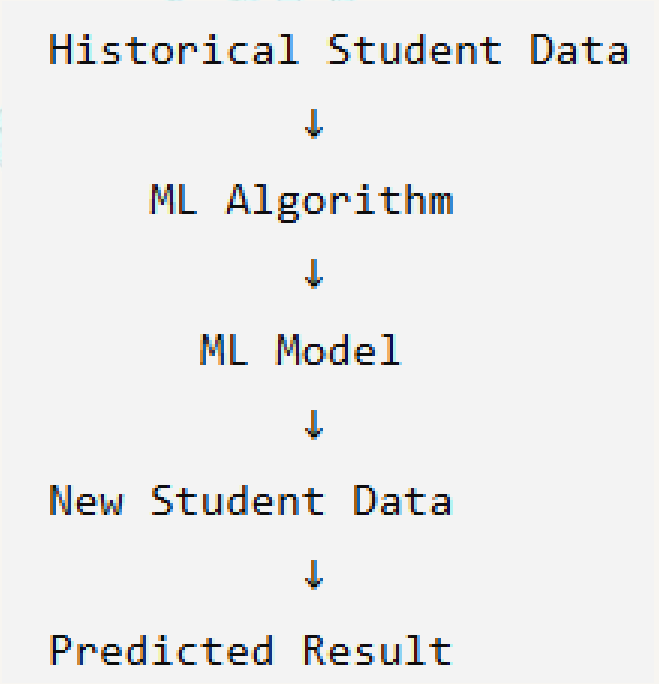
AI vs Conventional Programming

Machine Learning

➢A machine-learning system can instead use:

- Data + Expected Outputs → Learning Algorithm → Model
- Then
- New Data + Model → Prediction

➢For example:



Prepared By: Mohammad Rasheed

Goals/Capabilities of Artificial Intelligence

➢AI attempts to develop systems capable of:

1. **Perception**
   - Understanding information from the environment.
   - Examples
     - Face recognition
     - Objection detection
     - Speech recognition

Prepared By: Ehtisham Rasheed

Goals of Artificial Intelligence

2. Learning

- Improving performance based on data or experience
- Examples:
  - Recommendation systems
  - Fraud detection
  - Predictive models

3. Reasoning

- Drawing conclusions from available information
- Example:

```
All humans are mortal.
Ali is human.
Therefore Ali is mortal.
```

Prepared By: Ehtisham Rasheed

Goals of Artificial Intelligence

4. **Problem solving**
   - Finding solutions to complex problems
   - Examples:
     - Route planning
     - Scheduling
     - Puzzle solving
5. **Decision making**
   - Choosing an appropriate action
   - Examples:
     - Medical decision support
     - Autonomous vehicles
     - Financial systems

Prepared By: Ehtisham Rasheed

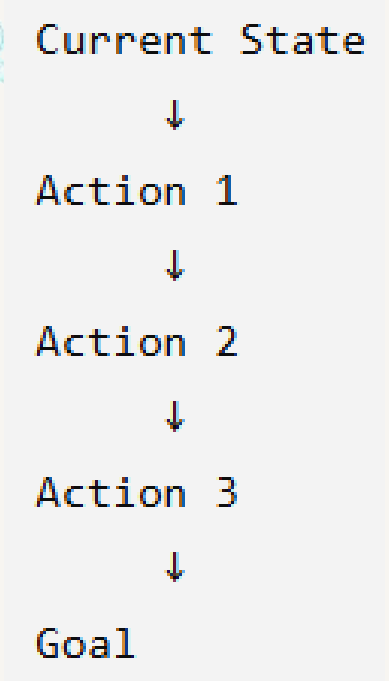
Goals of Artificial Intelligence

6. Natural Language Understanding

- Processing human language
- Examples:
  - Chatbots
  - Translation
  - Sentiment analysis
  - Question answering

7. Planning

- Determining a sequence of actions to achieve a goal
- Example:



Prepared By: Ehtisham Rasheed

Characteristics of an Intelligent System

➢An intelligent system generally demonstrates some combination of:

Characteristic	Meaning
Perception	Obtains information from the environment
Reasoning	Uses information to draw conclusions
Learning	Improves from data/experience
Decision Making	Selects appropriate actions
Problem Solving	Finds solutions to problems
Planning	Determines future actions
Adaptation	Adjusts to changing situations
Autonomy	Performs tasks with limited human intervention

➢Not every AI system necessarily has all these characteristics

Real-World Applications of AI

- **Healthcare**
- AI is used for:
  - Medical image analysis
  - Disease prediction
  - Patient monitoring
  - Drug discovery
  - Clinical decision support
- **Finance**
- Applications include:
  - Fraud detection
  - Credit scoring
  - Risk assessment
  - Algorithmic trading
  - Customer service

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Real-World Applications of AI

- **E-Commerce**
- Examples:
  - Product recommendations
  - Customer behavior prediction
  - Personalized advertising
  - Chatbots
  - Demand forecasting
- **Transportation**
- AI is used in:
  - Route optimization
  - Traffic prediction
  - Driver assistance
  - Autonomous vehicles
  - Fleet management

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Real-World Applications of AI

- **Education**
- Examples:
  - Personalized learning
  - Automated assessment
  - Student performance prediction
  - Intelligent tutoring system
  - AI assistants
- **Manufacturing**
- AI can support:
  - Predictive maintenance
  - Quality inspection
  - Robotics
  - Production optimization
  - Demand forecasting

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Real-World Applications of AI

- **Cybersecurity**
- AI can help identify
  - Suspicious activity
  - Malware
  - Network anomalies
  - Fraudulent behavior
  - Intrusion patterns

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Major Areas of Artificial Intelligence

- Students will encounter following major AI areas in this course
- **Search and Problem Solving**
  - Finding solutions efficiently
  - Examples:
    - A*
    - BFS
    - DFS
- **Knowledge Representation**
  - Representing knowledge in a form a computer can reason about
  - Examples:
    - Logic
    - Rules
    - Knowledge graphs

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Major Areas of Artificial Intelligence

➢Machine Learning

- Systems learn patterns from data
- Examples:
  - Classification
  - Regression
  - Clustering

➢Deep Learning

- Uses multi-layer neural networks
- Examples:
  - Image recognition
  - Speech recognition
  - NLP

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Major Areas of Artificial Intelligence

➢Natural Language Processing

- Computers process human language
- Examples:
  - Chatbots
  - Translation
  - Sentiment analysis

➢Computer Vision

- Computers understand images and video
- Examples:
  - Face recognition
  - Object detection
  - Medical imaging

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Major Areas of Artificial Intelligence

➢Rototics

- AI combined with physical machines
- Examples:
  - Industrial robots
  - Autonomous robots
  - Service robots

➢Generative AI

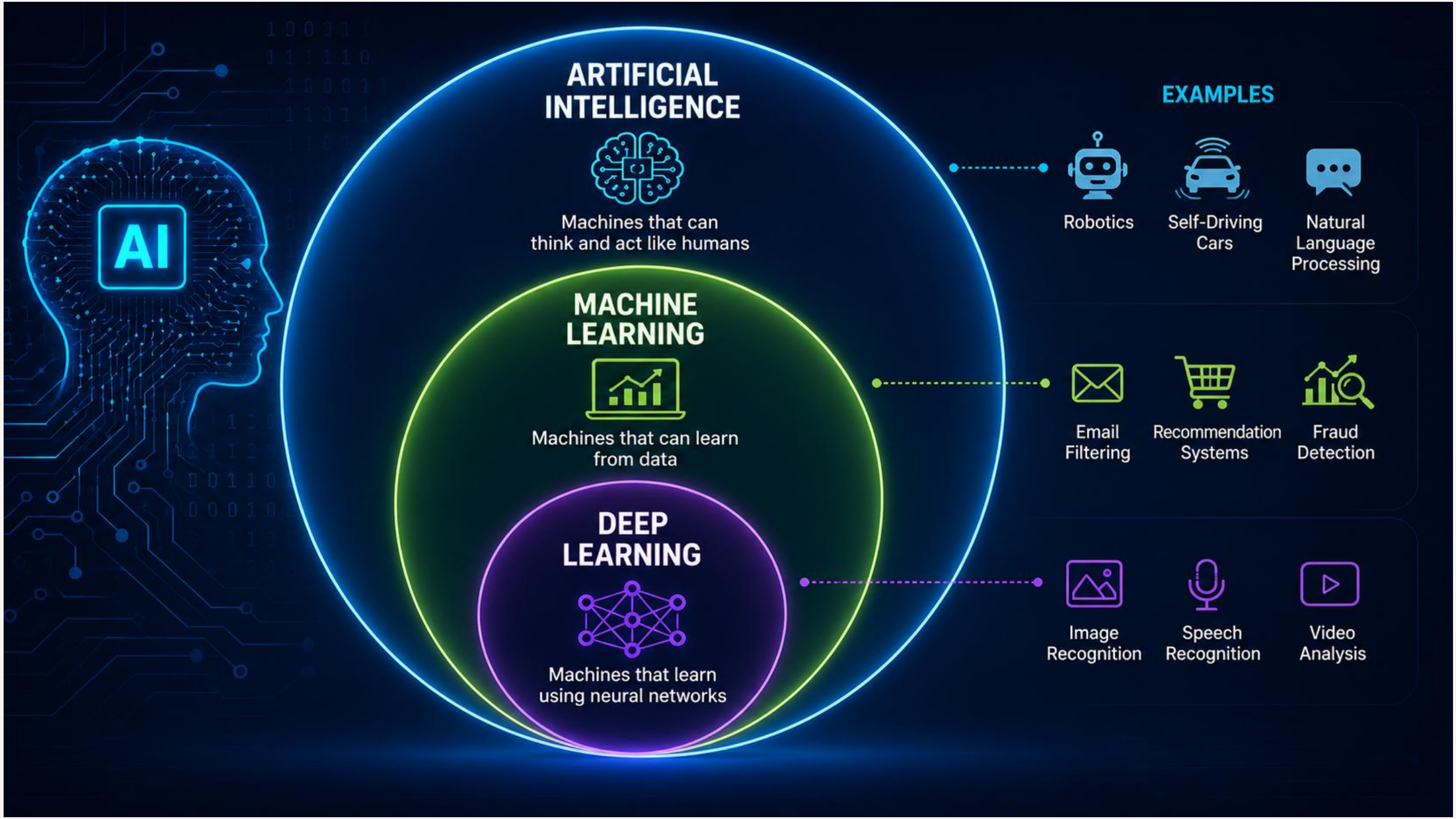
- Systems generate new content
- Examples:
  - Text, Images
  - Audio, Video
  - Code

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AI, Machine Learning and Deep Learning

- Artificial Intelligence
  - It is a broad field
  - Goal: Build systems capable of intelligent behavior
- Machine Learning
  - A major approach within AI
  - Goal: Allow systems to learn patterns from data
- Deep Learning
  - A family of machine-learning techniques
  - Uses: Multi-layer neural networks to learn complex patterns

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AI
ARTIFICIAL INTELLIGENCE
Machines that can think and act like humans
MACHINE LEARNING
Machines that can learn from data
DEEP LEARNING
Machines that learn using neural networks
EXAMPLES
Robotics
Self-Driving Cars
Natural Language Processing
Email Filtering
Recommendation Systems
Fraud Detection
Image Recognition
Speech Recognition
Video Analysis

Simple Example

- Suppose we want to recognize cats and dogs
- Traditional Programming

```
IF ears are pointed
AND body has certain shape
AND ...
THEN CAT
```

The problem is that real-world images vary enormously

Machine Learning

```
10,000 Cat Images
+
10,000 Dog Images
        ↓
Machine Learning
        ↓
Trained Model
```

Then provide:

```
New Image
    ↓
Model
    ↓
CAT
```

Prepared By: Fa ashed

Types of AI

1. Narrow AI / Weak AI

- Designed for a specific task or limited set of tasks
- Examples
  - Spam detection
  - Recommendation systems
  - Face recognition
  - Chess AI
  - Voice recognition
- Almost all practical AI systems today fall into this category

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Types of AI

2. General AI / AGI (Artificial General Intelligence)

- A hypothetical system capable of performing a broad range of intellectual tasks at a human-like level
- It would not be limited to one particular task
- Examples
  - Learn Programming + Learn Mathematics + Understand Language + Solve New Problems + Learn New Skills
- AGI is a research concept

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Types of AI

3. ASI (Artificial Super Intelligence)

- This refers to a hypothetical intelligence that would substantially exceed human intellectual capabilities
- This is primarily a theoretical concept and is important when discussing the future

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AI Development Cycle

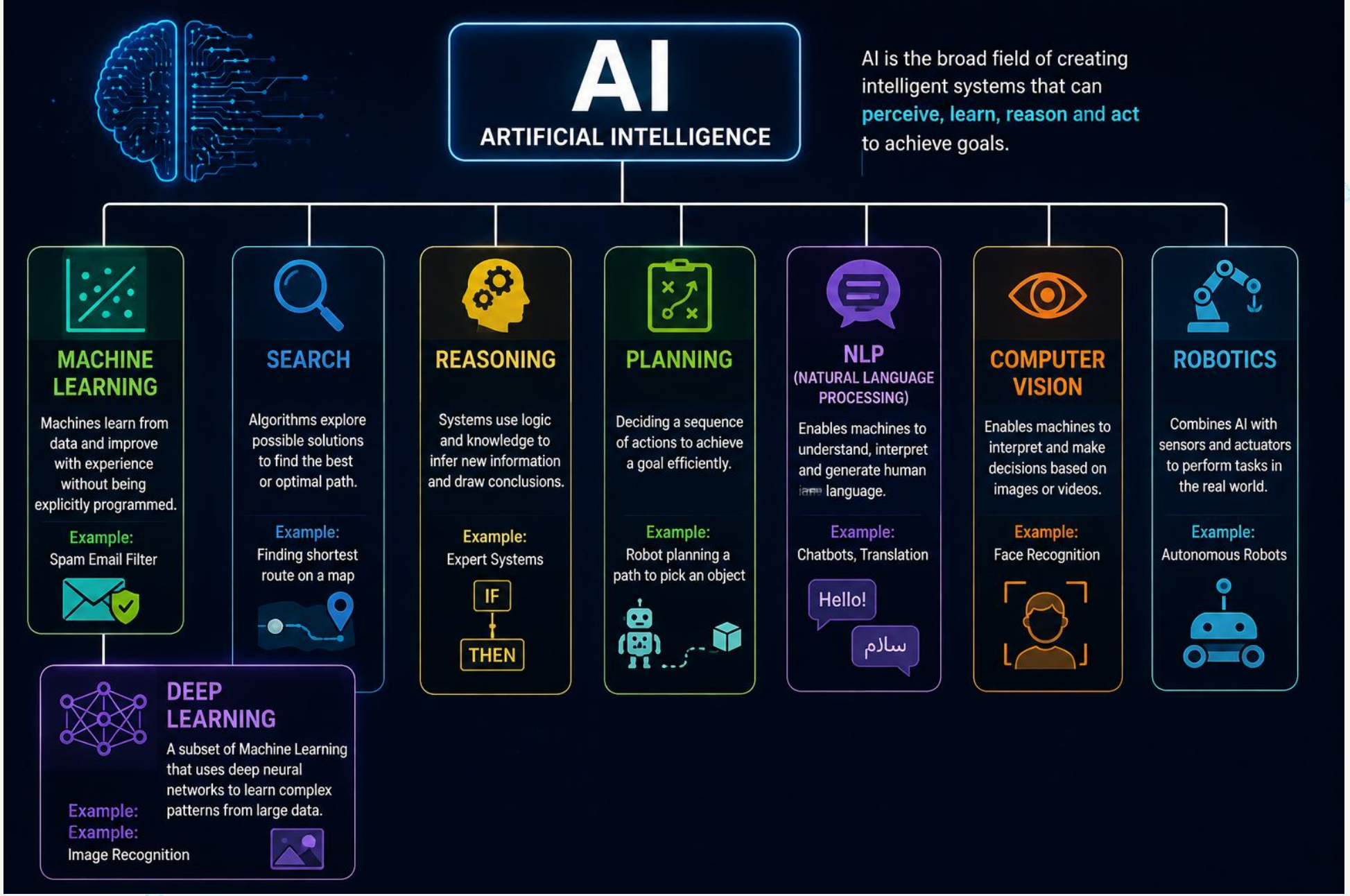
- Step 1 — Define the Problem
  - What are we trying to solve?
- Step 2 — Collect Data
  - What information is available?
- Step 3 — Prepare Data
  - Clean and transform the data
- Step 4 — Select an Approach
  - Search?
  - Rule-based reasoning?
  - Machine learning?
  - Neural network?

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AI Development Cycle

- Step 5 — Build the System
  - Implement the solution
- Step 6 — Evaluate
  - Measure performance
- Step 7 — Improve
  - Modify the system based on results
- Step 8 — Deploy
  - Use the system in a real environment

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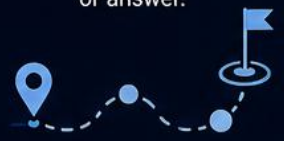
This image is not saying that all these are completely independent branches.
There is significant overlap shown in next slide

ARTIFICIAL INTELLIGENCE

Classical AI

Search

Algorithms that explore possible solutions to find the best path or answer.



Examples:
- BFS, DFS
- A* Search
- Best-First Search

Reasoning

Using logic and knowledge to infer new information and draw conclusions.

All humans are mortal.
Ali is a human.
Therefore, Ali is mortal.

Examples:
- Logic
- Inference
- Knowledge Representation

Machine Learning

Deep Learning

Uses deep neural networks to learn complex patterns from large amounts of data.



Examples:
- Image Recognition
- Speech Recognition
- Generative AI

Robotics

Vision

Enables machines to perceive and understand the visual world.



Examples:
- Object Detection
- Face Recognition
- Scene Understanding

Planning

Deciding the best sequence of actions to achieve a goal.



Examples:
- Path Planning
- Task Scheduling
- Robot Navigation

NLP (Natural Language Processing)

Enables machines to understand, interpret and generate human language.

Examples:
- Chatbots
- Translation
- Text Summarization

Computer Vision

Enables machines to interpret and make decisions from images or videos.

Examples:
- Image Classification
- Object Detection
- Medical Imaging

AI Does Not Always Mean Machine Learning

- A common misconception is:
  - "AI = Machine Learning."
  - This is incorrect.
- For Example:
  - A* search
    - A* is a classical AI algorithm
    - It does not necessarily learn from data
  - Minimax
    - Minimax is a classical AI algorithm for game playing
  - Rule-Based Expert System
    - A system can use predefined rules

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Classical AI vs Conventional Programming

- **Conventional Programming** means we explicitly write the rules/steps that the computer should follow

```
if marks >= 50:
    result = "Pass"
else:
    result = "Fail"
```

- **Classical AI** (also called **Symbolic AI**) refers to AI approaches that explicitly represent knowledge and use logical rules, search, reasoning, planning, etc
- Examples:
  - A* search, Minimax, Expert systems, Knowledge representation, Planning

Prepared By: Ehti...

Classical AI vs Modern AI

- Classical / Symbolic AI often focus on:
  - Rules, logic, knowledge representation, search, planning, reasoning
- Modern Data-Driven AI often focus on:
  - Data, statistical patterns, machine learning, neural networks, deep learning
  - Modern AI often combines multiple approaches

Prepared By: Ehtisham Rasheed

Advantages of AI

- Automation
  - AI can automate repetitive tasks
- Speed
  - AI can process large amounts of data quickly
- Pattern Recognition
  - AI can identify patterns that may be difficult for humans to detect manually
- Availability
  - Software systems can operate continuously
- Decision Support
  - AI can help humans make informed decisions
- Personalization
  - AI can provide personalized recommendations and experiences

Prepared By: Ehtisham Rasheed

Limitations of AI

- Data Dependency
  - Many AI systems require large quantities of useful data
- Bias
  - If training data contains bias, an AI system may learn or reproduce that bias
- Lack of Explainability
  - Some complex AI models are difficult to interpret
- Errors
  - AI systems can produce incorrect results
- Computational Requirements
  - Some AI models require significant computational resources
- Privacy Issues
  - AI applications may process sensitive or personal information

Prepared By: Ehtisham Rasheed